



Figure 6: Mount folder of NFS sever



Figure 7: Partition selection menu

or USB pen drive. Note that you may need to configure your BIOS before you can boot from these devices. Upon booting, you will get a screen as shown in Figure 1.

Select the first option 'Clonezilla live'. On the next screen, you will get to choose your default language. On the following screen, select the 'Start_Clonezilla' option. On the next screen, you will get the menu to select 'local dev', 'ssh server', 'samba server', 'nfs server' or 'enter shell' as shown in Figure 2.

Select 'nfs_server' on this screen. On the following screen, select the 'device-image' option as shown in Figure 3.

On the following screen, select the network you will use, as shown in Figure 4.

Depending on your network configuration, you can choose *dhcp* or *static* to configure your Clonezilla on the network. This is very straightforward. In some cases, you may need to contact the systems administrator to set the values for the gateway and mask. Home users can leave the default values and go ahead with the next step. After correctly setting up the network, Clonezilla will ask to mount the NFS server directory on to the local directory as shown in Figure 5 and Figure 6. In our case, the *nfs* directory is '/mnt/backup'.

This *nfs* directory is mapped to */home/partimag*. In the next screen, select the *Beginner* option. Depending on your requirements, you can choose 'saveparts' or the 'savedisk' option. In this tutorial, I will choose only Windows partitions to back up, as shown in Figure 7. The * indicates that the particular partition is selected for back-up.

In the screens that follow, you can choose default options as necessary. It asks for final confirmation before continuing with the backup. On this screen, you can cross check the configuration and restart the procedure, if that is what you want to do. Clonezilla verifies the health of the backup and whether there is a possibility of corruption on its own, hence ensuring that the backup is clean and usable. Finally, choose the *Power off* option to exit Clonezilla, as shown in Figure 8.

This completes the back-up steps.

Restore Windows partition(s)

If there is any adverse situation that requires you to

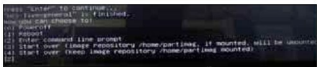


Figure 8: Power off



Figure 9: Clonezilla back-up and restore options



Figure 10: Choose target partitions

restore this partition or deploy this back-up to the homogeneous systems over the network, you can use the *Restore* option of Clonezilla. Follow all the steps of the previous section up to the point of mounting the NFS server directory to the local *Clonezilla/home/partimag* directory. After this step, Clonezilla will automatically recognise that this directory has a previously saved back-up and it will show the *restoreparts* and *restoreparts* options (Figure 9). At this point, you can choose to create a new back-up or restore the old one.

To restore the back-up, choose *restoreparts*. On the next screen, Clonezilla will ask you to choose the name of the backup, in case you have more than one. Select the latest back-up that you want to restore. On the next screen, it will ask you to choose the target partitions that you want to restore. Be sure to choose the same partition that you made the backup of, or else Windows will not be able to boot. This is shown in Figure 10.

After this step, select all the default options and continue with the restore. And after it is completed, you can safely reboot the system. This completes the restore procedure.

In this article, we have explored a simple way to back up and restore Windows partitions over NFS server. However, the steps mentioned are general and applicable to Samba servers also. One needs to know the Samba server address and configuration. 

By: Saurabh Jha

The author is an open source enthusiast who loves to create awareness about free and open source software and technologies. His interest includes high performance computing in heterogeneous environments. He can be reached at his website <http://saurabhjha.in>